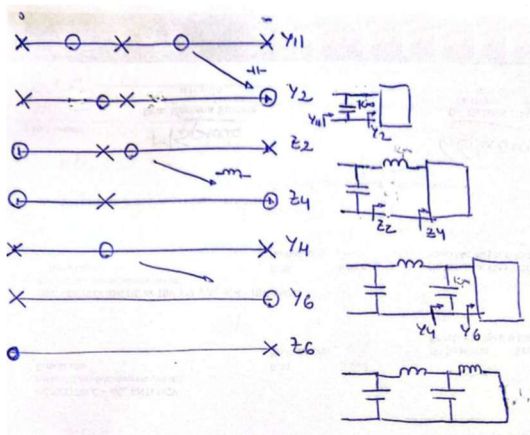


Ej 3

$$y_{21} = \frac{1}{s(s^2+4)} \quad y_{11} = \frac{(s^2+1)(s^2+9)}{s(s^2+4)} \quad \frac{I_2}{I_1} = \frac{y_{21}}{y_{11}} = \frac{\frac{1}{s(s^2+4)}}{\frac{(s^2+1)(s^2+9)}{s(s^2+4)}}$$



Calculo residuos:

$$k_{\infty} = \lim_{s \rightarrow \infty} \frac{y_{11}}{s} = 1$$

$$y_2 = y_{11} - s = \frac{s^4 + 10s^2 + 9}{s^3 + 4s} - s = \frac{6s^2 + 9}{s^3 + 4s}$$

$$z_2 = \frac{s^3 + 4s}{6s^2 + 9}$$

$$k_{\infty} = \lim_{s \rightarrow \infty} \frac{z_2}{s} = \lim_{s \rightarrow \infty} \frac{s^2 + 4}{6s^2 + 9} = \frac{1}{6}$$

$$z_4 = z_2 - \frac{1}{6}s = \frac{s^3 + 4s - \frac{1}{6}s(6s^2 + 9)}{6s^2 + 9} = \frac{4s - \frac{3}{2}s}{6s^2 + 9}$$

$$z_4 = \frac{5/2 s}{6s^2 + 9}$$

$$y_4 = \frac{6s^2 + 9}{5/2 s}$$

$$k_{\infty} = \lim_{s \rightarrow \infty} \frac{y_4}{s} = \lim_{s \rightarrow \infty} \frac{6s^2 + 9}{5/2 s^2} = \frac{12}{5}$$

$$y_6 = y_4 - \frac{12}{5}s = \frac{6s^2 + 9 - \frac{12}{5}s \cdot \frac{5}{2}s}{\frac{5}{2}s} = \frac{18}{5s}$$

$$z_6 = \frac{5}{18}s$$

